Industrial Safety Processes

Process Safety Management

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OSHA Process Safety Management

"….the proactive identification, evaluation, and mitigation or prevention of chemical releases that could occur as a result of failures in processes, procedures, or equipment."

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., p. 718.

Process Safety Management

- Focus on high-hazard, high-risk industrial operations
- Petrochemical, mining (primarily coal), industrial chemical
- Primarily managed by technical specialists with associated education – i.e., engineers, industrial chemists

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Process Safety Focus

- Identifying potential problems that could lead to large-scale or catastrophic failures
- Putting in place control measures to prevent or mitigate the consequences of potential problems
- Strong focus on system / plant / operational integrity to the exclusion of almost every other consideration

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Importance for OHS Professional

- Involvement in safety-by-design processes usually involve some element of process safety management
- Involves examination of systems / processes safety considerations – catastrophic events, system incidents
- Distinct from “personal safety” issues (focus on injury or worker wellbeing as highest priority)
- Expands perspective and utility as an OHS professional

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Process Safety

1. Higher energy systems
2. Large scale, severe events
3. Focus on engineering & design of systems

Generalist OHS

1. All energy levels
2. All events
3. Focus on safety of persons interacting with systems

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Kerin, T. (2019). Managing Process Safety. In The Core Body of Knowledge for Generalist OHS Professionals. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 4.

Elements of PSM Programs

- Process safety information
- Employee involvement, communication & training
- Process hazard analysis
- Management of change
- Operating procedures
- Work practices & permits
- Contractor management
- Pre-start up reviews
- Quality assurance
- Maintenance
- Emergency preparedness
- Compliance auditing
- Investigations
- Standards & regulations

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Hagan, Philip E., Montgomery, John F., & O'Reilly, James T., eds. (2015). *Accident Prevention Manual for Business & Industry: Engineering & Technology* (14th Edition). Itasca, Ill.: National Safety Council., pp. 722-729.

Chemical Process Hazards

- Hazardous substances
  - Flammable, explosive, oxidizing, corrosive, compressed gases, reactive substances
- Sources of ignition
  - Flame, hot surfaces, mechanical friction, electrostatic discharge, electrical equipment, vehicles
- Operating temperatures
  - Low and high temperatures

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, pp. 5-14.

Combined Gas Law

"If the temperature changes then either the pressure or volume or both will change in proportion to the temperature…

"Most importantly in process safety, as the volume decreases the pressure increases, and as the pressure for a fixed volume of gas increases, the temperature will also increase."

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 4.

Energy

"Energy cannot be created or destroyed, but it can be transferred to another object or transformed into another form of energy."

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 4.

Chemical Reactions

"Chemical reactions involve the breaking and/or making of molecular bonds with associated changes in energy level. Breaking molecular bonds requires an input of energy (endothermic reaction) while creating bonds releases energy (exothermic reaction)."

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 5.

“The rate of a chemical reaction is affected by:

- Concentration and physical state of the reactants
- Temperature
- Surface area of the reactants
- Presence of solvents and catalysts;
- And, for gases, pressure”

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 5.

Approaches to Process Safety Analysis

- Concept hazard analysis
- Hazard identification (HazID)
- Process hazard review (PHR)
- Hazard & Operability Study (HAZOP)
- Consequence (dispersion) modelling
- Fault tree
- Event tree
- Quantitative risk assessment (QRA)
- Layers of protection analysis (LOPA)
- Safety integrity level (SIL) analysis
- Failure modes and effects analysis (FMEA)
- Societal risk (FN curve)

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Kerin, T. (2019). Managing Process Safety. In The Core Body of Knowledge for Generalist OHS Professionals. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, pp. 36-38.

Process Safety Incident

"… an event involving the release/loss of containment of hazardous materials or energy that can result in large-scale health, safety, environmental consequences"

CSA Z767-17 Process Safety Management

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Process Safety Incident Consequences

- Fires
- Explosions
- Toxic Effects

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, pp. 14-23.

American Petroleum Institute (API)

API RP-754 *Process Safety Performance Indicators for Refining and Petrochemical Industries*

- Tier 1 – Greater Consequence
- Tier 2 – Lesser Consequence
- Tier 3 – Challenges to Safety Systems
- Tier 4 – Operating Discipline & Management System Performance Indicators

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Tier 1 – Greater Consequence

- Loss of Primary Containment (LOPC)
- Large release of hazardous substances
- Lost time days and fatalities
- Community evacuations
- Shelter in place orders
- Incident damage exceeding 100,000

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Tier 2 – Lesser Consequence

- Loss of Primary Containment (LOPC)
- Smaller release of hazardous substances
- Indicator of problem with process safety system
- Potential for
  - Recordable injuries
  - Incident costs over 2500

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Tier 3 – Challenges to Safety Systems

- Loss of Primary Containment (LOPC) of minor extent
  - Small leak, emissions
- Indicator of control or system failure
  - Near miss incidents

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Tier 4 – Operating Discipline & Management System

- Audit findings
- Failure to implement or follow process safety system:
  - Incomplete or missed inspections
  - Incomplete or missed maintenance
  - Training, communication, etc.

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The International Association of Oil and Gas Producers (IOGP) – Hardware Barriers

- Structural integrity
- Process containment
- Ignition control
- Detection systems
- Shutdown systems
- Protection systems, including deluge and firewater systems
- Emergency response
- Life-saving equipment, including evacuation systems.

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 26.

Passive (Engineering Controls)

"Passive barriers are the most reliable and are usually developed at the design stage. They do not require action by a person or device; they perform their function just by existing, but often require maintenance to ensure they continue to meet the design performance standards. Examples include:

- Making equipment more robust (e.g. appropriate thickness of materials)
- Shielding by fire walls and doors, explosion walls
- Collapsible panels (e.g. explosion vents for buildings with combustible dusts, entire walls for gas compression buildings)."

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Active (Engineering & Administrative)

"Active barriers detect a hazardous condition and initiate an automated response. Such equipment includes:

- Alarms and trip systems
- Pressure relief valves
- Automated shutdown systems
- Automated deluge systems.."

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Administrative or Procedural

"Administrative or procedural barriers, which are the least reliable as they involve people carrying out specific tasks, may include:

- Operating procedures
- Operator-initiated process adjustments
- Operator-initiated response to alarms
- Muster and evacuation procedures."

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Kerin , LT (2019). Process Hazards (Chemical). In *The Core Body of Knowledge for Generalist OHS Professionals*. 2nd Ed. Tullamarine, VIC: Australian Institute of Health and Safety, p. 27.